Geospatial USD: proposal quality and working evidence

Two newly written placement readers pass 19 contract groups against the unchanged proposal.

Its existing rules were sufficient to correct the implementation defects found during the run. No new blocking specification gap was found in these workflows, and no undocumented geospatial fields were needed. This is bounded evidence, not proof of exhaustive conformance.

Store inputs; compute results

A coordinate reference system (CRS) gives numbers their location, axes, units and height reference. USD stores that definition as well-known text (WKT), alongside the author's placement decisions. Projection effects and resolved positions are computed results; they do not become extra source USD properties.

Authored input	Meaning
crs:wkt	One complete CRS definition, shared through composition
crs:position	Absolute model origin in that definition's components and units
crs:orientation	Physical heading, pitch and roll against geodetic east/north/up
Ordinary xformOps	Model-local descendants; project adjustments on the anchor
External-data association	Asset, format, selected field and coordinate domain

The codeless schema resources contain the proposed input fields.

Placement has an explicit order

Child transforms first produce model-local geometry. Source position and physical attitude place it on the Earth. The anchor's ordinary scale, rotation and translation then act in a fixed working chart. Changing the requested output CRS preserves that adjustment instead of interpreting the same numbers along different axes.

Stage of evaluation	What it preserves
Child transforms and asset conformance	Ordinary reusable model-local meaning
Source position and geodetic attitude	Physical origin, direction and local lengths
Conversion into the working chart	A fixed context for the anchor's adjustments
Anchor's complete ordinary stack	Authored scales, pivots, rotations and translations
Requested coordinate/scene output	Adjusted physical placement in the requested representation

Reset stacks, independently bound children, reference offsets, inherits, value clips and native instance proxies are exercised.

Provider examples check the coordinate interpretation

Across 123 comparison rows, both readers use equivalent realized operations and agree with the supplied provider tables within the predeclared 2 mm distance. The largest provider residual is 0.100 mm. These distances test interpretation and implementation agreement; they do not establish survey accuracy.

Conversion	Rows	Largest provider residual
France: geographic to geocentric	59	0.100 mm
France: CC49 to Lambert-93	59	0.061 mm
Colorado: State Plane to calibrated site	5	0.034 mm

Provider tables also use PROJ; these checks do not establish independent geodetic or survey accuracy.

The same Eiffel geometry reaches ordinary USD consumers

From authored location to a checked placement

1 Author the placement

WGS 84 geographic 3D

2.2945° E 48.8584° N

Ellipsoidal height 70 m

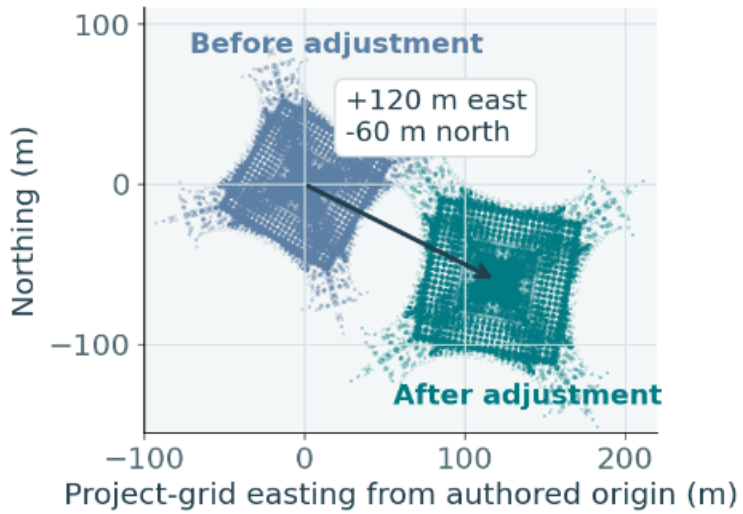
Heading 30° from true north



300 m model after asset conformance

2 See the adjustment in metres

Plan view of the actual tower vertices



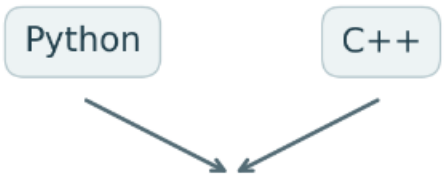
Ordinary USD: scale 1.15×, rotate 20°, then translate

3 Compare the two results

163,416 vertices checked

≤ 0.016 mm

largest Python / C++ difference



Compare every vertex

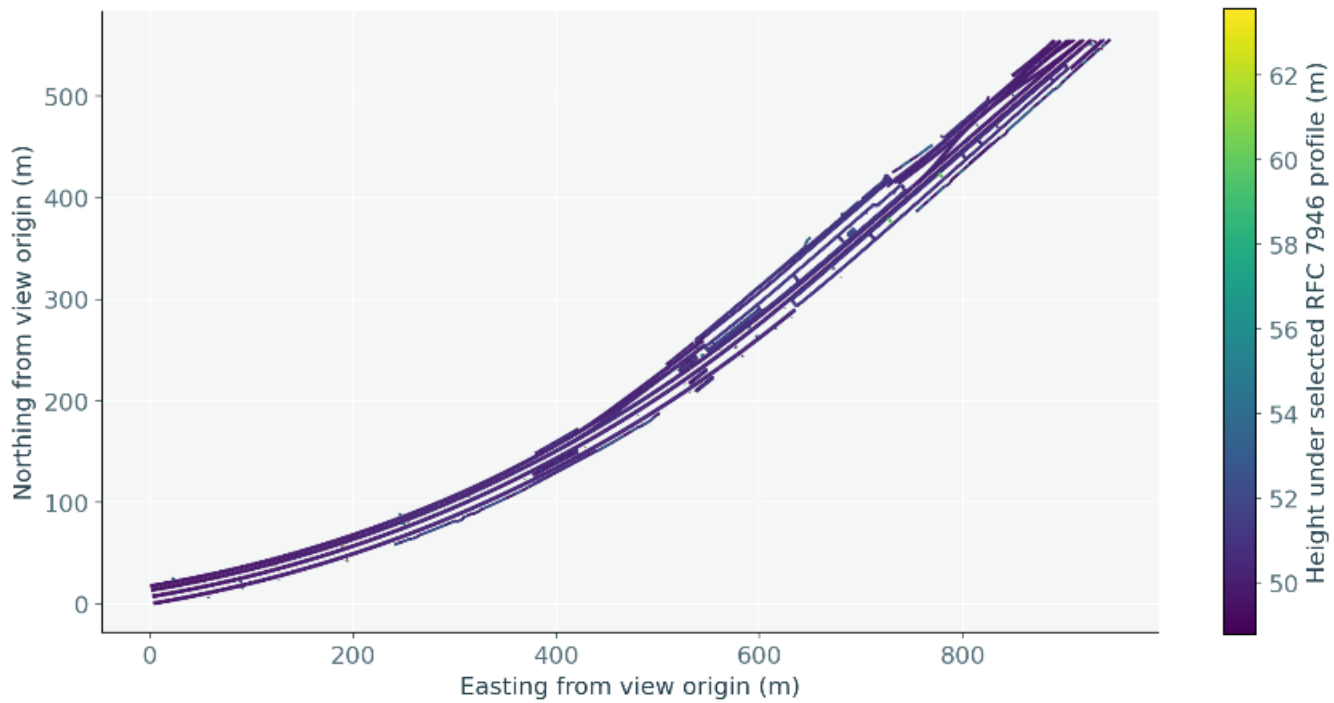
Ordinary USD outputs reach
Hydra and Omniverse consumers.

Illustrative placement, not surveyed Paris context. Both implementations use PROJ; agreement is not independent geodetic validation.
Tower geometry: SDC PERFORMANCE, (FREE) La tour Eiffel, CC BY 4.0. Size and placement modified.

Authored placement, then USD adjustment. Tower: SDC PERFORMANCE, CC BY 4.0 (modified). Both readers use PROJ.

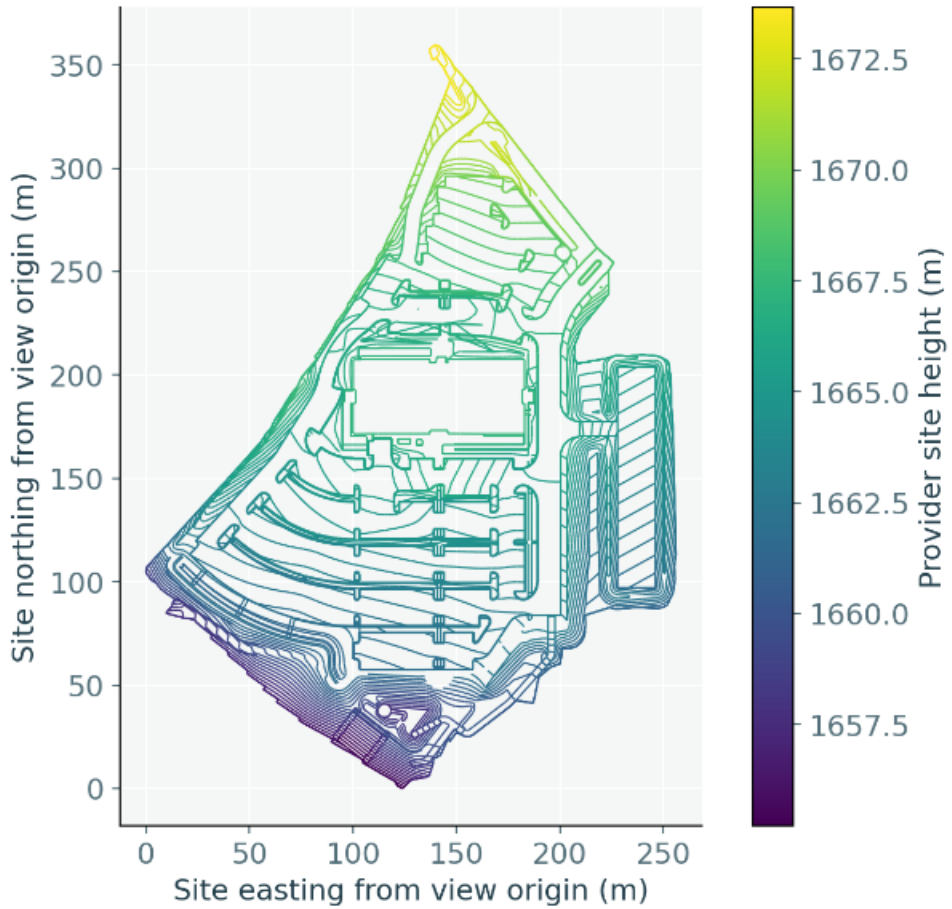
The original railway remains useful as data

Both readers resolve the same 15,822 railway positions and retain feature/geometry indices. The visualization uses those positions and preserves separate line sequences. A consumer can obtain coordinates and feature values without constructing renderable geometry, then ask for another supported output CRS.



The selected RFC 7946 association defines height interpretation; the legacy source labels horizontal WGS 84.

Site calibration survives the data exchange



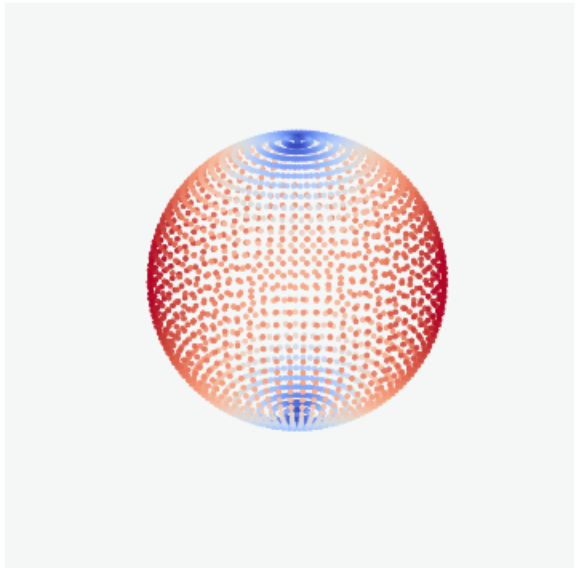
The original Westminster data contributes 14,359 polyline positions, including 13,899 breakline points. Both readers retain the calibrated site's full WKT and US survey feet while converting the same records to the State Plane context. Horizontal and vertical calibration remain CRS meaning, separate from object placement.

A query client supplies LandXML records and companion WKT; no standard LandXML USD association is claimed.

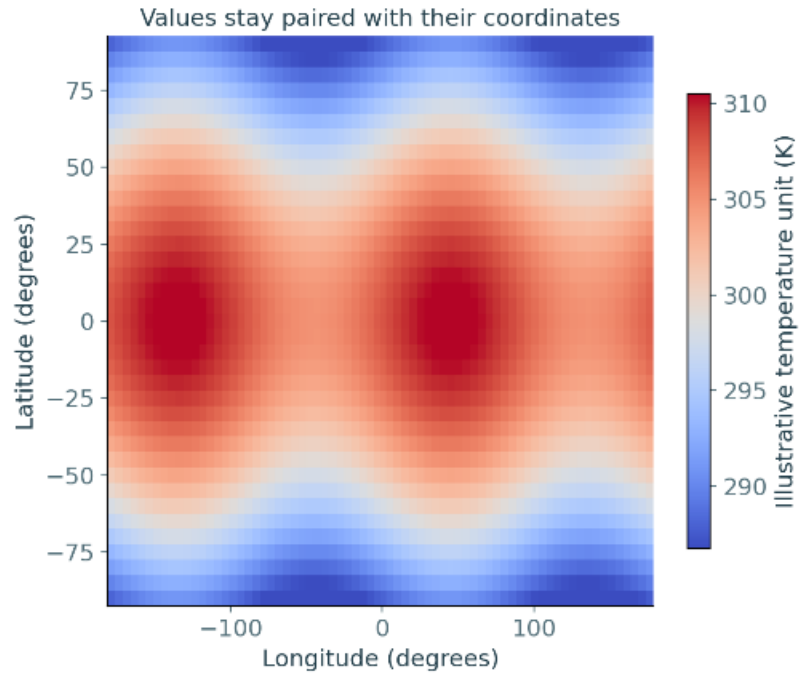
Global analysis does not require a renderer

The climate example pairs 2,664 original measurement values with explicit coordinates and two illustrative observation times, yielding 5,328 sample records. Both readers agree on their global geocentric locations. Changing a measurement time keeps its location association intact; the globe and heatmap are optional views of the same data.

The same samples in a global Cartesian frame



Values stay paired with their coordinates



Added example inputs: Kelvin, zero ellipsoidal height, chosen observation times and a second field at +1 K.

A height fallback is visible and queryable

When a model placement lacks a vertical CRS, the specified fallback is EGM2008 gravity-related height, with a warning. At the illustrative Eiffel coordinate, 35 m in that reference resolves to about 79.639 m ellipsoidal height. The reader reports the source interpretation and whether it was declared or assumed; it authors no computed height properties.

Source definition	Computed height report
Geographic 3D with ellipsoidal height	Ellipsoidal; declared
Geocentric X, Y, Z	No height axis; declared
Horizontal geographic CRS used for model placement	EGM2008 height in metres; assumed and warned
Horizontal projected CRS used for model placement	EGM2008 height in the horizontal linear unit; assumed and warned

Required grids are actually used. A missing grid fails visibly instead of substituting another height reference.

Profiles lets a consumer detect the dependency early

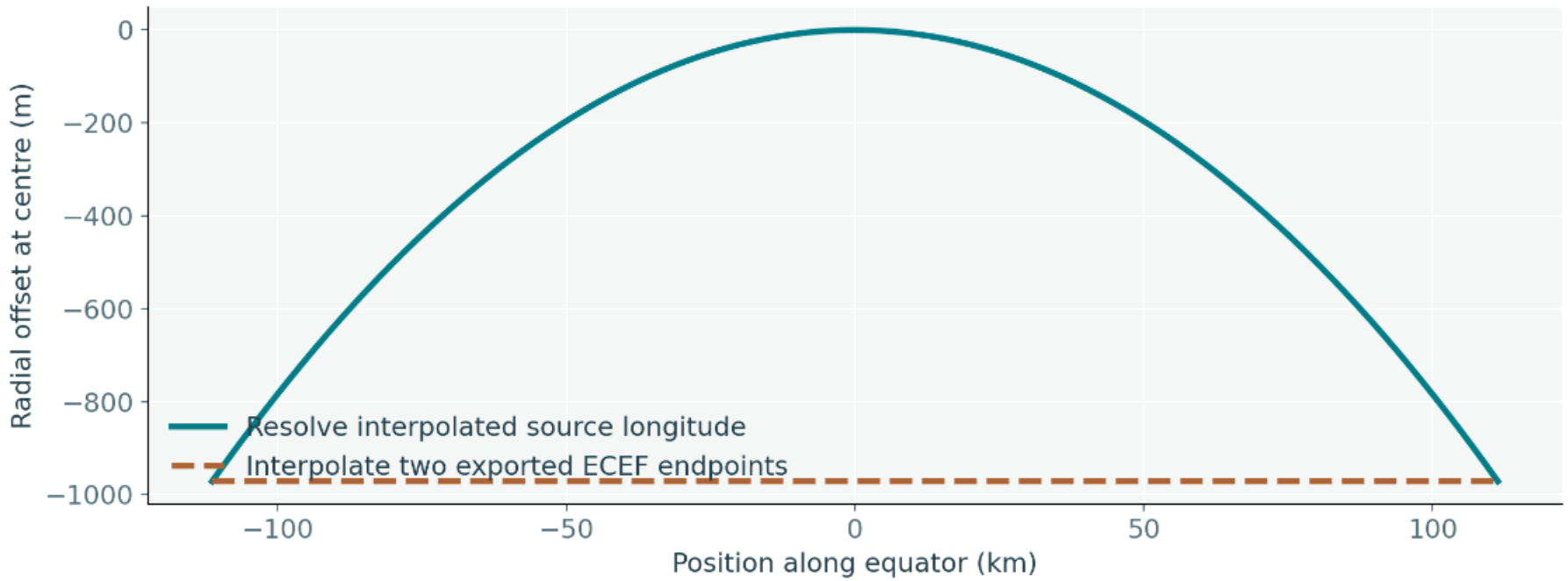
Authors use the geospatial input schemas. Publishers maintain a conservative summary at advertised entrypoints, above payloads and across retained variants. Consumers read that summary before loading content: hard dependency, qualified known-none, or unknown. This helps adoption by making unsupported placement detectable instead of silently wrong.

Participant	Observable contract
Author	Binding and measurement schemas imply the proposed capability
Publisher	ClaimsAPI records a complete, conservative publication summary
Consumer	Reads the summary without traversing or loading payloads
Ordinary geometry bake	May claim known-none after every retained dependency is removed

Native ClaimsAPI reads the declaration with zero payloads loaded. The capability names remain proposed.

Sampling and coverage claims have practical limits

A two-endpoint ECEF bake does not preserve the source trajectory between those endpoints. The equatorial test exposes a 971.421 m midpoint difference. Exported sample keys and time-code scale record what was written; they do not claim an untested path or certify the continuous image of a face, curve or subdivision surface.



The sampled bake claims its actual sample instants; it makes no between-sample trajectory guarantee.

Separate proposal quality from prototype coverage

The fresh run found implementation defects and repaired them against existing requirements. It found no new blocking model or runtime ambiguity in the exercised contracts. Remaining review/adoption questions, explicit roadmap deferrals and prototype coverage limits are different claims and must remain visible when assessing readiness.

Category	Current assessment
Missing meaning in tested contracts	No new blocking gap found; bounded evidence
Review and adoption	Detailed candidate conventions and Profiles identity ownership still require standards review
Explicit deferrals	Coordinate epochs and scene-authored operation/resource pinning
Prototype limits	General continuous-domain certification, custom GeoTIFF decoding and point-instancer bake expansion are visibly unsupported
Independence of evidence	Same author and PROJ family; provider tables also use PROJ

Reverse audit.